Distance in Stan v2

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1 Introduction

Here we demonstrate a model-based Bayesian distance sampling analysis using an HMC algorithm, implemented using Stan via the `R` package `rstan`. Thanks to

- Juan Morales for his online example code (from 2020, in an Rpubs document), which provided some initial inspiration;
- the excellent Stan Function Reference and Users Guide;
- MMARINeDNA collaborators for helpful suggestions;
- ChatGPT for providing some further suggestions for improvements.

Currently, we fit data using a half-normal detection unction (with no adjustment terms) and use this to estimate probability of detection. We combine with encounter rate to estimate density – but without including uncertainty on that component – this will come next!

We test the function on some simulated data, and on datasets available in the `R` `Distance` and `inlabru` packages. We compare results with those derived using maximum likelihood from the `Distance` package.

2 Background

2.1 Distance sampling detection function likelihood

A canonical reference for basic distance sampling is Buckland et al. (2001); here we give a very compact summary of relevant components. In line transect distance sampling, observers traverse a set of linear transects searching for objects of interest (animals or animal groups). We assume that the probability of detecting objects depends on their distance from the transect line. This is described using a "detection function", $g(x)$, which gives the probability of detection of an animal at distance x from the line. Parametric or semi-parametric functions are used for g. These are generally set up under the assumption that $g(0) = 1$ – i.e., that the probability of detecting animals at 0 distance is 1. (There are lots of further options, such as having g depend on additional variables such as observer, weather, etc., and allowing $g(0) < 1$, but we ignore them here.) Observations further away than some specified "truncation distance", w are generally discarded, and hence $g(x)$ for $x > w$ is 0.

Under a further assumption that true (observed and unobserved) animal distances are uniformly distributed wrt distance from the transect line (which is reasonable given random line placement within the study area), then pdf of observed animal distances, $f(x)$ is given by

$$f(x) = \frac{g(x)}{\int_0^w g(x)dx}. \tag{1}$$

The integral in the denominator is typically called the "effective strip width" (ESW; technically it is the effective strip half-width) and denoted μ. (Dividing this quantity by the truncation distance gives the average probability of detecting an animal that is within distance w of the line – i.e., $p = \mu/w$.)

In conventional distance sampling, the detection function is estimated using maximum likelihood. Given a set of n detections with observed distances $\mathbf{x} = (x_1, \dots, x_n)$ the log-likelihood of detection function parameters θ is given by

$$\begin{aligned} \ell(\theta; n, \mathbf{x}) &= \sum_{i=1}^{n} \log(f(x)) \\ &= \sum_{i=1}^{n} \log(g(x)) - n\log(\mu). \end{aligned} \tag{2}$$

(Note that here we are treating the number of detections n as fixed; clearly there is information in this number about the number of animals but we ignore this for the present as we are just focussing on estimating the detection function.)

2.2 Half-normal detection function

Here we will work with a very simple parametric detection function, the "half-normal", which is given by

$$g(x) = \begin{cases} \exp\left(-\frac{x^2}{2\sigma^2}\right) & x \le w \\ 0 & x > w \end{cases} \tag{3}$$

where σ is the single parameter of this function. Given this detection function, the log-likelihood is given by

$$\ell(\sigma; n, \mathbf{x}) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i^2) - n\log(\mu). \tag{4}$$

Conveniently, μ in this case can be evaluated without the need for numerical integration, as

$$\mu = \sigma\sqrt{\pi/2} \times \mathrm{erf}\left[w/\left(\sigma\sqrt{2}\right)\right] \tag{5}$$

where $\mathrm{erf}(x)$ denotes the error function evaluated at x.

3 Bayesian model

Our goal here was to develop a Bayesian distance sampling model, and fit it using the Stan software.

One consideration is the prior distribution for σ. After some experimentation, we specify a normal prior on $\log\sigma$. This is discussed further in Section 3.2.

3.1 Stan code

Below is the developed Stan model code.

```
//uncomment the following line, and the corresponding one at the bottom
// to run from inside R (e.g., RStudio)
//cds_hn_stan <- stan_model(model_code = "
data {
  int n; //Number of observations
  vector[n] x; //Vector of observed distances
  real w; //Truncation distance
  real log_sigma_prior_mean; //Mean of normal prior on log sigma
  real<lower=0> log_sigma_prior_sd; //SD of normal prior on log sigma
}
parameters {
  real log_sigma; //Log of half normal sigma parameter
}
transformed parameters {
  real<lower=0> sigma = exp(log_sigma); //Define sigma
  //Calculate effective strip width - can be done analytically in this case
  // as the detection function is a half normal with no adjustments
  real esw_hn = sigma * sqrt(pi() / 2) * erf(w / (sqrt(2) * sigma));
}
model {
  //Priors
  log_sigma ~ normal(log_sigma_prior_mean, log_sigma_prior_sd);
  //Likelihood - vectorized
  target += -sum(square(x)) / (2.0 * square(sigma)) - n * log(esw_hn);
}
generated quantities {
  real p = esw_hn / w;
}
//")
//uncomment the above to run from inside R
```

3.2 Prior hyperparameters and prior sensitivity

As noted previously, we specify a normal prior on log σ. This was the result of some experimentation. We initially tried a flat prior, such a a uniform, or the conjugate inverse gamma on $1/\sigma^2$ with small α and β values). This was in order to most closely conform with the MLE approach that is currently the standard method. However, we found (a) the NUTS sampler in Stan tended to give some divergent transitions, and (b) estimates of σ tended to have a long right tail, particularly for smaller datasets. This makes sense because once σ gets much larger than about $2w$, the resulting detection function is flat, and increasing values of σ produce detection functions with practically indistinguishable likelihoods. This is illustrated in Figure 1.

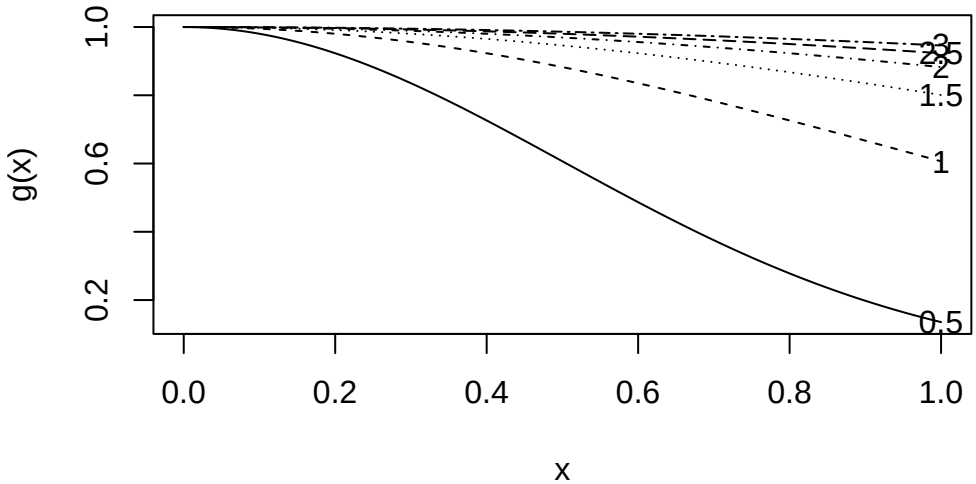

Figure 1: Example half-normal detection functions, illustrating how the function flattens out as σ increases. In this example the truncation distance, w is set to 1. σ values used are shown on the rhs of the plot.

We therefore switched to specifying a normal prior on $\log \sigma$. With suitable choice of hyperparameters, this allowed us to specify a gently decreasing density with increasing σ at larger values of σ. Another advantage was that this distribution gives low densities to very small values of σ, which are not plausible in practice as they would denote a very "spiked" detection function, with probability of detection going to small values at relatively small distances. If such a situation were found in practice, a smaller value of w would be used in the analysis. The value of w used is often an analysis choice; for example Buckland et al. (2001) recommend truncating at a value of w where $g(w) \approx 0.15$.

A sensible choice of prior mean on $\log \sigma$, which we will denote $\mu_{l\sigma}$, will depend on the scale of the distance data. Since a sensible w also depends on the scale of the data, it makes intuitive sense for the prior mean to be set by the value of w. Indeed, if the analyst follows the advice to have $g(w) \approx 0.15$ then this suggests a value to use, since we can back-calculate a σ from Equation 3:

$$\begin{aligned} g(x) &= \exp\left(-\frac{x^2}{2\sigma^2}\right) \\ 0.15 &= \exp\left(-\frac{w^2}{2\sigma^2}\right) \\ \sigma &= \frac{w}{\sqrt{-2\log(0.15)}} \\ &= w/1.9479 \\ &\approx w/2 \end{aligned} \tag{6}$$

Hence, a reasonable prior mean parameter for log σ is $\mu_{l\sigma} = \log(w/2)$. (Note that this parameter for a lognormal distribution corresponds to the median on the natural scale.)

In choosing the prior standard deviation on log σ, which we will denote $\sigma_{l\sigma}$ we want a value that has the desirable properties of mildly discouraging very small values of σ, or values greater than approximately $2w$, while not overly influencing the result (e.g., posterior mean on σ) with small datasets. Note that Buckland et al. (2001) recommend a minimum sample size of approximately 60 observations for line transect analysis, so in this context a "small" dataset will be one of about that size. The standard deviation of a lognormal distribution is approximately equal to the coefficient of variation (CV) of the distribution – this relationship holds for small CVs; the exact relationship is $\sigma^2_{l\sigma} = \log(1 + \mathrm{CV}^2)$ and hence $\mathrm{CV} = \sqrt{\exp(\sigma^2_{l\sigma}) - 1}$. Values in the range of 0.5 to 1 seem reasonable, since we would hope that a sample of data would considerably improve a prior CV of these values (although 0.5 may perhaps be a bit too informative). We undertook a small sensitivity analysis using value of $\sigma_{l\sigma} = 0.5$, 0.8 and 1.0 (corresponding to CVs of 0.53, 0.95 and 1.31) and report findings in the Results section. For reference, the pdf of these three prior distributions on $\log \sigma$ is shown in Figure 2.

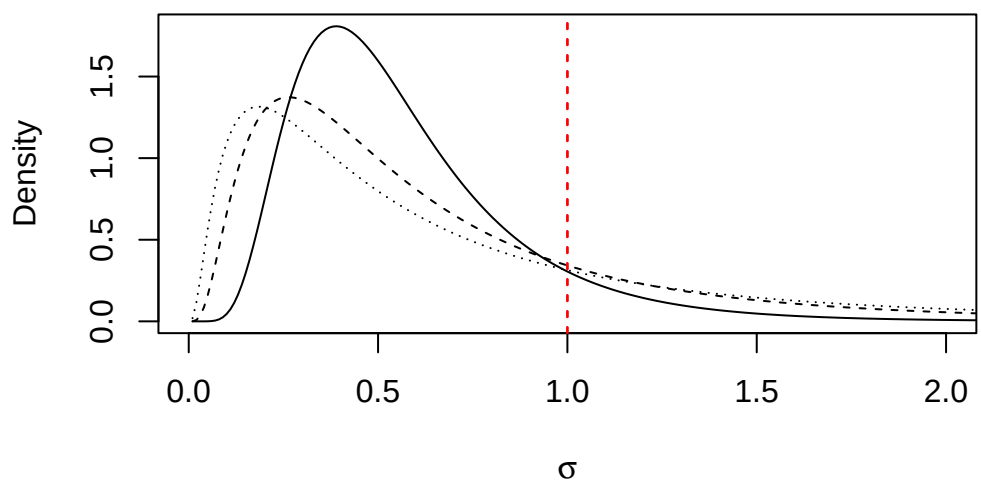

Figure 2: Three potential prior distributions for log σ (transformed to the natural scale to show the implied prior on σ). All specify a normal distribution on log σ, and all have the same lognormal mean parameter of 0.5 (i.e., $w/2$ given a w in this example of 1, shown by the vertical red dashed line). The solid, dashed and dotted lines have lognormal sd parameters of 0.5, 0.8 and 1.0 respectively.

Also implies a prior distribution on average detection probability p, which we can calculate.

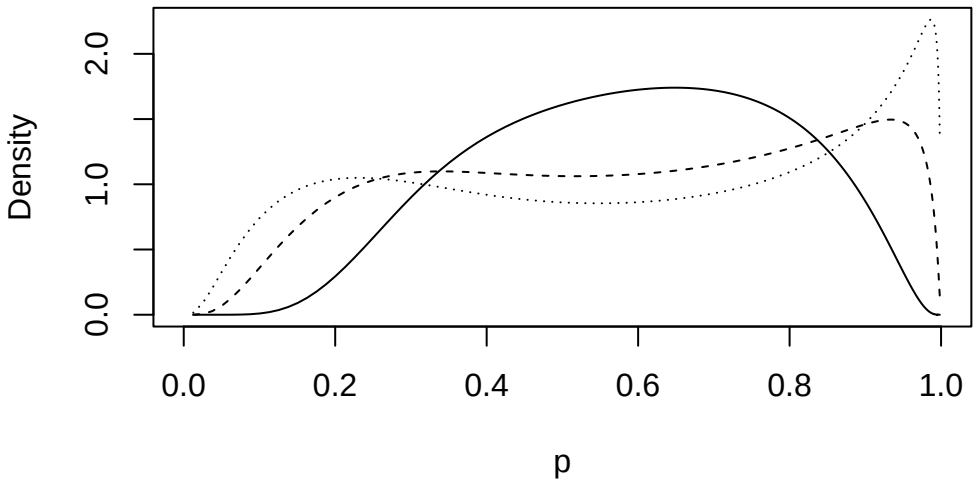

Figure 3: The implied prior on p (average detection probability between 0 and w of the three potential prior distributions for log σ. All three specify a normal distribution on log σ, and all have the same lognormal mean parameter of 0.5 (i.e., $w/2$ given a w in this example of 1, shown by the vertical red dashed line). The solid, dashed and dotted lines have lognormal sd parameters of 0.5, 0.8 and 1.0 respectively.

4 Datasets

4.1 Simulated data

We initially fitted the model to a simulated dataset of simulate 10^4 samples from a half-normal detection function with $\sigma = 1$ and truncation distance $w = 10$. The reason to have such a large truncation distance is that there is an analytical formula for the maximum likelihood estimate of the half-normal sigma parameter from untruncated data (Buckland et al. 2001, section 3.3.4.1):

$$\hat{\sigma}^2 = \sum_{i=1}^{n} x_i^2/n \tag{7}$$

We can therefore check the result we get from Stan against this – given the size of the dataset we expect the MLE to be very close to the posterior mean, for example (although we could also check the posterior mode if desired). We note, however, that with this value of w we are very far from the advice that $g(w) \approx 0.15$ which motivated our choice of prior mean for $\log \sigma$.

Nevertheless, this dataset is really just a proof that the code is not obviously wrong.

4.2 GoMx Dolphin data

We next used a line transect dataset collected using visual observations from a shipboard survey in the Gulf of Mexico (GoMx) – see Miller et al. (2013). Here we use observations of pantropical spotted dolphin (*Stenella attenuata*); this dataset is provided with the `inlabru` package (and also available in `dsm`). Figure 4 shows the sample and detection locations and Figure 5 gives a histogram of the observed distances. There are 47 observations. The truncation distance used in previous analyses of these data is 1 m.

This dataset is a good example of one with a small sample size, so we expect more prior sensitivity.

4.3 Other datasets

We also tried the code on some other line transect datasets available in the `Distance` package (and the `mrds` package, which is a dependency of `Distance` and is the one that actually does the fitting – `Distance` is largely a wrapper for `mrds`). These datasets were:

- Wren. Songbird data collected by Steve Buckland near his (old) home in Fife, Scotland – see Buckland (2006).
- Sika deer. Data from a survey of Scottish sika deer dung pellets (sika deer spend most of their time in woodland areas, so abundance estimates are based on pellet group counts). These data are from the Peebleshire portion of the survey - see Marques et al. (2001).
- Golf tee. These data are from a survey of groups of golf tees laid out in a simulation exercise and surveyed by a student class at St Andrews using double-observer distance sampling methods (see Borchers et al. 2002). Here we just use the data from the first observation team.

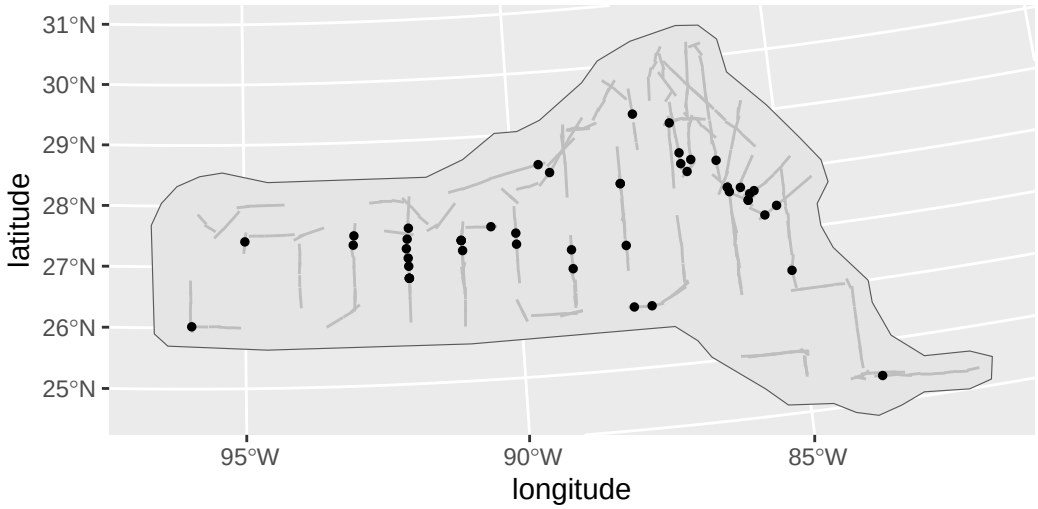

Figure 4: Location of transects (lines) and observations (points) in Gulf of Mexico dolphin data

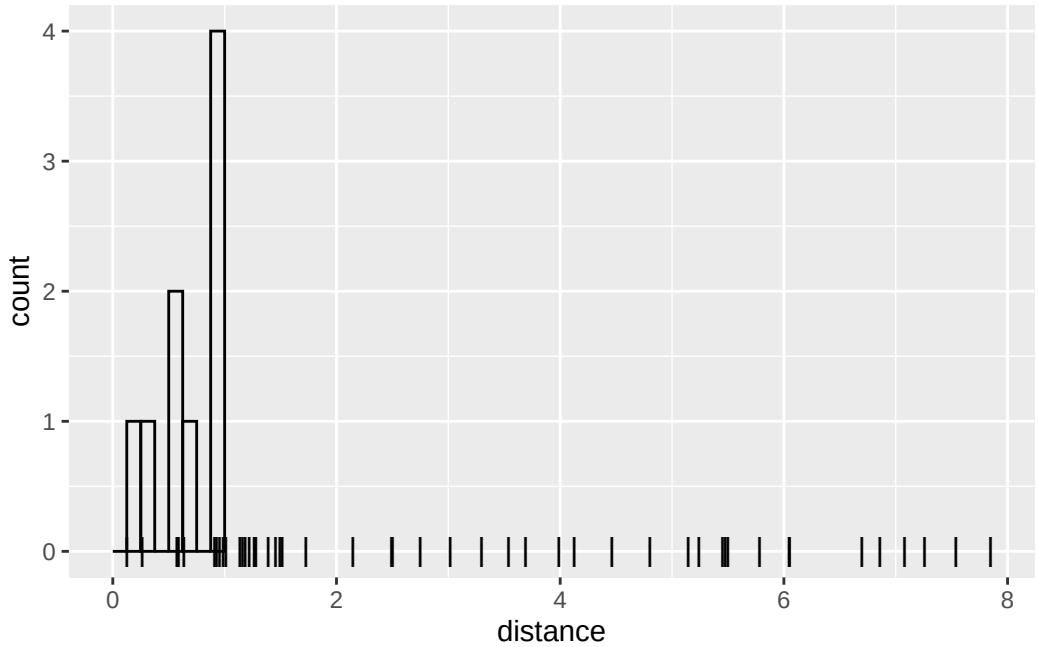

Figure 5: Histogram of detection distances. Individual detection distances are marked as tick marks on the x-axis.

- ETP dolphin. These data are from shipboard line transect surveys in the Eastern Tropical Pacific; the data here are for spotted dolphin schools – see Marques et al. (2003). Here we use only the data with "Cue type 3", which means data where the observers were cued by the presence of birds above the dolphin school. The reason for this is that this cue type has a slightly more "spiked" histogram of distances and in that sense is a bit different from the other datasets analyzed here.

A summary of the datasets is given in Table 1 and histograms of detection function distances are in Figure 6.

Table 1: Summary of datasets used in this report. n is the sample size of detections, w is the truncation distance. The last column is the median observed distance divided by w; lower values indicate more observations at smaller distances.

Name	n	w	median distance / w
GoMx dolphin	47	8	0.313
wren	156	100	0.350
deer	1731	175	0.371
golf tee	124	4	0.318
ETP dolphin	307	5	0.300

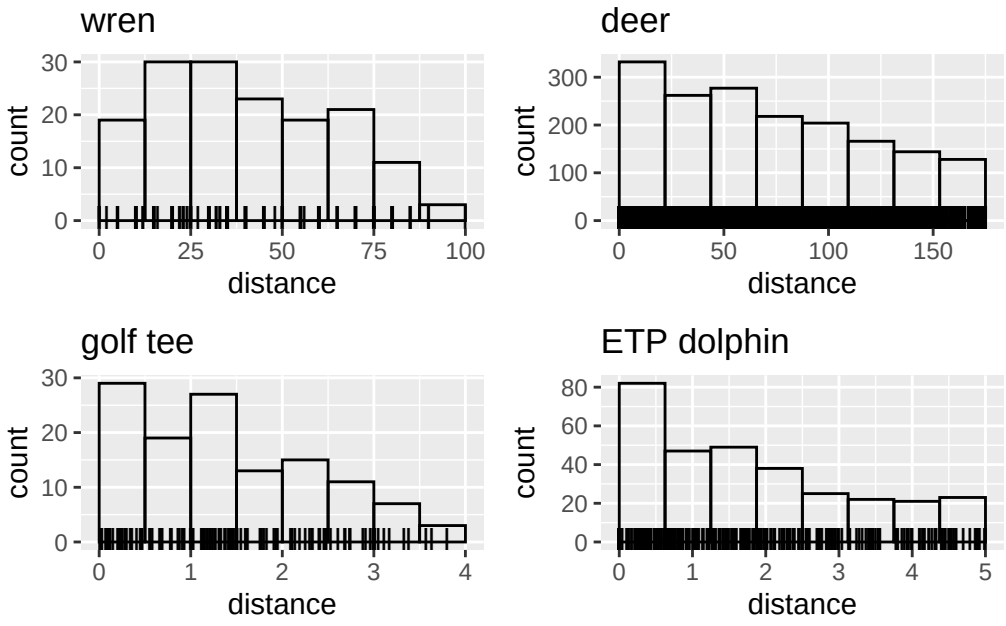

Figure 6: Histogram of detection distances for each of the additional datasets used. Individual detection distances are marked as tick marks on the x-axis.

5 Results

All Stan model results reported are based on 6 chains and 3000 samples per chain (half of which were discarded as warmup), with no thinning. Monte Carlo error on reported results was not checked formally, but is likely to be small. Start values for σ were chosen at random, sampling uniformly between $w/100$ and $2w$. During Stan runs, we monitored for diagnostic warnings and did not receive any – note that Stan messages are suppressed from the output in this rendered file, so it would be necessary to run the code in RStudio interactively to see any warning messages.

5.1 Simulated data

Analytic evaluation of the MLE using Equation 7 gives 1.0006931, very close to the true value of 1 used to simulate the data. (Recall that this does not take truncation into account, but we do not expect it to make any difference in this case as all distance are very far from the truncation distance.)

We then fit using the `Distance` package (where truncation is taken into account). A summary of the output is given below. Note that the `N in covered region` on the last line is meaningless as we have not passed in any information about survey effort (line length covered).

```
Summary for distance analysis
Number of observations :  10000
Distance range         :  0  -  10

Model       : Half-normal key function
AIC         :  14531.68
Optimisation:  mrds (nlminb)

Detection function parameters
Scale coefficient(s):
                estimate         se
(Intercept) 0.0006928184 0.00709519

                       Estimate           SE          CV
Average p           1.254183e-01 8.898666e-04 0.007095191
N in covered region 7.973320e+04 9.359737e+02 0.011738820
```

The `(Intercept)` term is the log of the estimated sigma parameter. So, exponentiating we get $\hat{\sigma} = 1.0006931$.

Lastly we fit using Stan (in this case just using $\sigma_{l\sigma} = 0.5$ as we expect the prior to make a negligible difference to the results given the very large sample size of simulated distances). A summary of the Stan output is given below. Trace plots and a kernel is shown in Figure 7.

```
Inference for Stan model: anon_model.
6 chains, each with iter=3000; warmup=1500; thin=1;
post-warmup draws per chain=1500, total post-warmup draws=9000.

          mean se_mean   sd     2.5%      25%      50%      75%    97.5% n_eff
sigma     1.00    0.00 0.01     0.99     1.00     1.00     1.01     1.01  3272
p         0.13    0.00 0.00     0.12     0.12     0.13     0.13     0.13  3272
lp__  -7270.49    0.01 0.68 -7272.42 -7270.64 -7270.23 -7270.06 -7270.02  3754
      Rhat
sigma    1
p        1
lp__     1

Samples were drawn using NUTS(diag_e) at Sun Apr 26 14:59:17 2026.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).
```

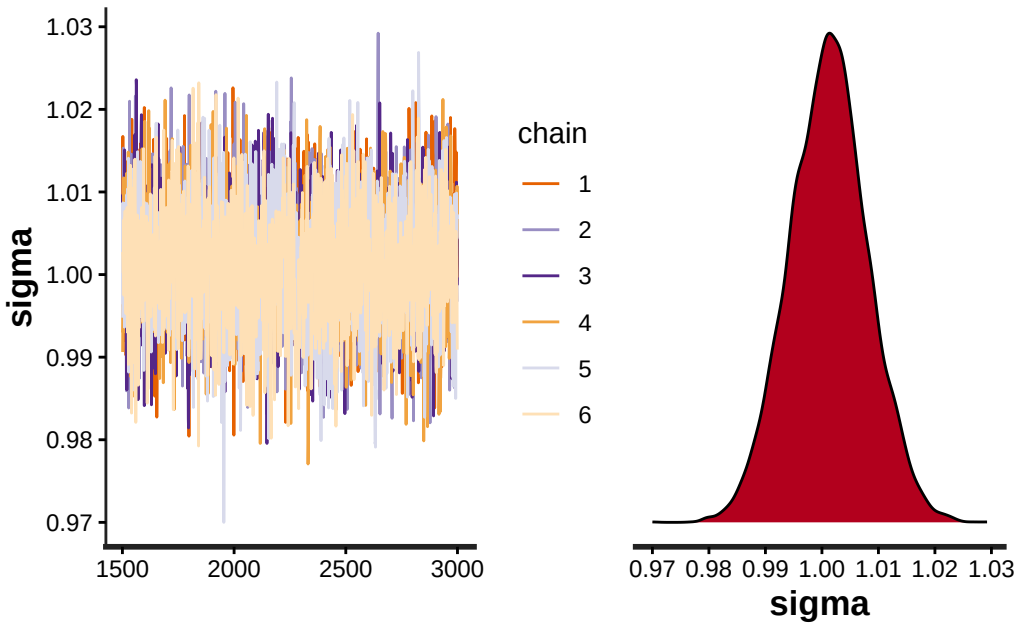

Figure 7: Trace plot (left) and density(right) for sigma from Stan run on simulated data.

The posterior mean for σ was 1.0012218 and the posterior mode was 1.0092669 – as expected, very close to the true value. We can also extract from the Stan outputs the value of p (average detection probability between 0 and w), as this was saved as a generated quantity. The posterior mean was 0.1254845, which compares well with the value estimated by `Distance` of 0.1254183. It is also of interest to compare measures of uncertainty. The SD of the samples from the Stan model run was 8.663955×10^{-4}, which is very similar to the SD on p (i.e., the SE) estimated by `Distance` (which uses the inverse Hessian evaluated

at the MLE) of 8.8986657×10^{-4}.

5.2 GoMx dolphin data

The output from `Distance` is shown below (again, ignore 'N in covered region').

```
Fitting half-normal key function

AIC= 192.985

No survey area information supplied, only estimating detection function.


Summary for distance analysis
Number of observations :  47
Distance range         :  0  -  8

Model       : Half-normal key function
AIC         :  192.9853
Optimisation:  mrds (nlminb)

Detection function parameters
Scale coefficient(s):
            estimate        se
(Intercept) 1.671953 0.2300667

                      Estimate         SE        CV
Average p            0.7230935 0.09200807 0.1272423
N in covered region 64.9985100 9.65883686 0.1486009
```

The estimated σ is 5.3225502. Estimated p is 0.7230935 with SE of 0.0920081.

We ran Stan with each of the three priors on σ, i.e., with $\sigma_{l\sigma}$ of 0.5, 0.8 and 1.0. Results are shown in Table 2.

Table 2: Summary of Stan fits to the GoMx dolphin data

log sigma prior sd	mean sigma	mode sigma	sd sigma	mean p	mode p	sd p
0.5	5.7008	7.1649	1.4365	0.7303	0.7013	0.0822
0.8	6.6572	5.9339	3.2483	0.7602	0.6953	0.0984
1.0	7.5282	8.1777	6.3555	0.7696	0.5605	0.1057

Comparing the ML and Baysian results, it looks like the Bayesian posterior mean for σ and p are closer to the ML results when $\sigma_{l\sigma} = 0.5$ but that the standard deviation on p is a little lower in this case. This is

perhaps unsurprising as some information about σ has been given in the form of the prior. Mean σ and p increases as $\sigma_{l\sigma}$ increases, becoming further away from the MLE.

Figure Figure 8 of the fitted detection function

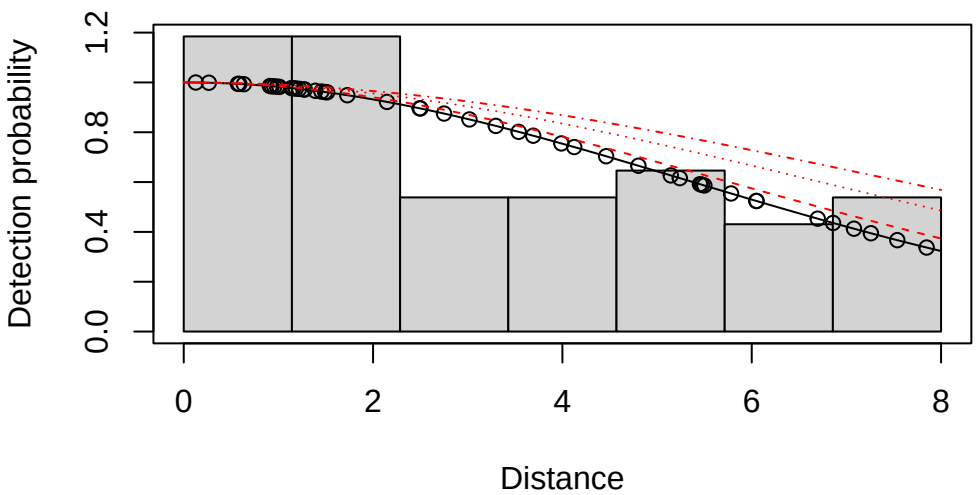

Figure 8: Fitted GoMx dolphin detection functions from Distance (i.e., maximum likelihood) (black) and from Stan (i.e., Bayesian)(red) with prior log sigma sd of 0.5 (dashed), 0.8 (dotted) and 1.0 (dot-dashed)

5.3 Other datasets

Results from the other datasets for σ and p are summarised in Table 3 and Table 4 respectively. The results generally concur with those found for the GoMx dolphins: posterior mean matches the MLE better with the more informative prior $\sigma_{l\sigma} = 0.5$, but the standard error of the estimate of p is lower than the MLE for this prior. One exception is ETP dolphins, where the posterior standard deviation was considerably larger than the standard error from the MLE at all levels of $\sigma_{l\sigma}$.

As expected, the posterior mean for σ and p are very similar to the MLEs for the dataset with the largest sample size, i.e., deer. In general the match scales with sample size and in any case differences between MLE estimate an posterior mean are small (see Table 3 and Table 4).

Table 3: Summary results for σ for other datasets

dataset	log sigma prior sd	MLE	posterior mean	posterior mode	posterior SD
wren	0.5	60.6923	62.8943	63.4079	8.2796
wren	0.8	60.6923	63.1959	53.3235	8.4845

Table 3: Summary results for σ for other datasets

dataset	log sigma prior sd	MLE	posterior mean	posterior mode	posterior SD
wren	1.0	60.6923	63.8409	63.1981	9.6417
deer	0.5	121.1189	121.4481	118.0192	5.3182
deer	0.8	121.1189	121.5886	116.6005	5.4535
deer	1.0	121.1189	121.8642	111.9788	5.5203
golf tee	0.5	1.9411	1.9979	1.7760	0.2109
golf tee	0.8	1.9411	1.9979	1.8138	0.2093
golf tee	1.0	1.9411	2.0026	1.9438	0.2164
ETP dolphin	0.5	2.7764	2.8106	2.6414	0.2080
ETP dolphin	0.8	2.7764	2.8187	2.8455	0.2146
ETP dolphin	1.0	2.7764	2.8162	2.7326	0.2118

Table 4: Summary results for p for other datasets

dataset	log sigma prior sd	MLE	MLE SE	posterior mean	posterior mode	posterior SD
wren	0.5	0.6850	0.0568	0.6944	0.7930	0.0501
wren	0.8	0.6850	0.0568	0.6962	0.6810	0.0513
wren	1.0	0.6850	0.0568	0.6991	0.7050	0.0545
deer	0.5	0.7386	0.0163	0.7390	0.7485	0.0167
deer	0.8	0.7386	0.0163	0.7394	0.7581	0.0170
deer	1.0	0.7386	0.0163	0.7402	0.7481	0.0172
golf tee	0.5	0.5843	0.0464	0.5947	0.6975	0.0463
golf tee	0.8	0.5843	0.0464	0.5948	0.6411	0.0464
golf tee	1.0	0.5843	0.0464	0.5956	0.5923	0.0478
ETP dolphin	0.5	0.6460	0.0296	0.6498	0.6790	0.0323
ETP dolphin	0.8	0.6460	0.0296	0.6509	0.6986	0.0329
ETP dolphin	1.0	0.6460	0.0296	0.6506	0.6145	0.0326

6 Discussion

We have demonstrated Bayesian estimation of the half-normal detection function for line transect analysis using Stan (accessed in `R` via `Rstan`). We suggested a normal prior on $\log \sigma$ as well as a suitable choice for the prior mean. FO rthe prior standard deviation ($\sigma_{l\sigma}$), we undertook a small sensitivity analysis, by trying the values 0.5, 0.8 and 1.0 on five datasets. Overall we found that $\sigma_{l\sigma} = 0.5$ gave the most similar estimate to the MLE, but that it could give a smaller estimated standard deviation of the posterior estimate of p. Differences were generally small, and overall we recommend using $\sigma_{l\sigma} = 0.5$.

If we ever write this up as part of a publication, it might be worth noting that the first Bayesian application to distance sampling we are aware of is Karunamuni and Quinn (1995), with another useful reference being Oedekoven (2014). A classic paper on weakly informative priors for variance parameters is Gelman (2006), although we would need to check whether it includes lognormal priors, and also whether it's in the context of random effects variances which is rather different from here (because here 0 is not a very likely number at all for the variance).

7 References

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